Household and Environmental Drivers of Gut Microbiome Variation in Ethiopian School Children
OR
Hygienic Habits and Household Variables Alter Gut Diversity and Composition in Ethiopian School Children

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1 Abstract

1.1 Background

The effects and associations between environmental, socioeconomic, clinical, and lifestyle factors and the gut microbiome remain understudied in non-western populations. Different living conditions and attitudes towards hygiene practices affect different populations differently. To explore this, household and environmental factors were tested for variations in the gut microbiome of 138 Ethiopian Schoolchildren.

1.2 Methods

Stool samples from 138 Ethiopian schoolchildren (aged 6-17) living in Jimma Town, Ethiopia, were collected and sequenced using 16s rRna targeted amplicon sequencing to identify the composition of the microbiome. The guardians of the children filled out the accompanying questionnaire. Statistical analysis of diversity and composition across socioeconomic, clinical, environmental, and household factors was performed in R.

1.3 Results

Lifestyle factors that were found to significantly alter gut diversity were handwashing soap usage after the toilet (chao1 $p = 0.02$; Shannon $p = 0.007$) school latrine use (chao1 $p = 0.04$), as well as demographic factors age (chao1 $p = 0.03$; Shannon $p = 0.08$) and weight (chao1 $p = 0.02$; Shannon $p = 0.08$). Kitchen floor material was found to alter gut composition ($p < 0.01$). Differences in abundance

of genera across variables was observed for handwashing soap usage (*Elusimicrobium* & *Prevotella*), school latrine use (*Blautia*, *Elusimicrobium*, & *Paludicola*), and kitchen floor material (*Lachnoclostridium* & *Prevotella_9*).

1.4 Conclusions

These exploratory identifications of non-physiological factors such as hand-washing habits and kitchen floor material influencing the gut microbiome can serve as references for future controlled experiments testing causal relationships.

2 Introduction

Study of the human gut-microbiome aims to characterize and analyze the relationship between the human host and the population of microorganisms inhabiting its gastrointestinal (GI) system. Primary interest is of the hundreds of billions of, mostly anaerobic, bacteria, unicellular eukaryotes, and other microorganisms in the human colon (Khalil et al., 2024). Humans and their microbiota form a symbiotic relationship that, for humans, extends the genomic capabilities of their GI system, aiding in digestion, immune function, metabolite synthesis, communicating with the nervous system through the gut-brain axis, and much more (Khalil et al., 2024).

Non-genetic factors account for roughly 20% of inter-human gut microbiome variation (Zhernakova et al., 2016). Common factors studied include diet, air pollutants, pet ownership, and smoking, as done in a large analysis of a Dutch population by Gacesa et al. African populations are underrepresented in human gut microbiome research, let alone for non-genetic factors(Ayeni et al., 2022). Most studies in this area focus on differences between urban and rural populations like in Nigeria (Ayeni et al., 2018), Tanzania (Pyuza et al., 2025), and South Africa (Ramaboli et al., 2024), instead of what underlying differences might be having an effect.

Household conditions vary across Africa and Ethiopia more specifically. In the Oromia region, containing Jimma Town, 12.6% of the population have access to improved toilet facilities, 61.5% to unimproved facilities, and 25.9% perform open defecation (Institute & Icf, 2021). This difference is exacerbated by wealth quintile (Institute & Icf, 2021). Microbes have been shown to be transmitted

through the air via substrates such as pollen and dust, and the biodiversity of the area can affect that transmission (Hanski et al., 2012).

The effects of hygienic habits on the gut microbiome are not well known. Monira et al. found that, in a small sample, a hand-washing program altered the abundance of certain gut bacteria relating to Cholera infection. Hand-washing is not ubiquitous in Ethiopian children despite knowledge of its benefits (Berhanu et al., 2022). Around half of children in Southern Ethiopia wash their hands before eating a meal, with most only using water instead of with soap (Berhanu et al., 2022). Hand-washing was more likely among older children, urban children, children with parents who were educators or health professionals, and unsurprisingly, those with more readily available access to hand-washing facilities (Berhanu et al., 2022).

In this cross-sectional study of 138 Ethiopian schoolchildren, we replicate previously found associations between gut microbiome diversity and age & weight, whilst also uncovering novel factors of gut microbiome composition such as soap use when hand washing and kitchen floor material.

3 Methods

Study methods have been articulated more precisely in previous studies [40]. Below is a more concise description of the methods.

3.1 Ethics Statement

The study was approved by the Institutional Review Boards (IRBs) of Colgate University, Hamilton, NY, USA (IRB Reference number: FR-S20-03/20) and Jimma University Institute of Health, Ethiopia (IRB Reference number: IHRPGD/786/20). Written or fingerprint consent was obtained for each child's parents or legal guardian. Children were informed of their right to leave the study at any point with no consequences. All confidential information on participating children has been kept in password-protected electronic files.

3.2 Study Design

The data presented in this study are one part of a larger, longitudinal & cross-sectional study analyzing the before-and-after effects of mass deworming protocols on the gut microbiome characteristics of Ethiopian Schoolchildren (Taye et al., 2024). Fecal samples were collected pre- and post-anti-helminthic treatment. Analyzed here are only the pre-treatment samples alongside personal information collected in the questionnaire regarding socioeconomic status, lifestyle habits, and environmental conditions.

3.3 Study Setting and Context

A detailed map was created for a previous study (Russel et al., 2023). Data was collected on children from ten primary schools in Jimma Town, Ethiopia. Jimma Town is a city of approximately 200,000 people (Service, 2007) and is 352 km from the capital city of Addis Ababa. The town sits 1.7km above sea level with a climate varying from dry to tropical. Jimma Town was chosen due to the prevalence of soil transmitted helminth infection in the area, the primary metric of another study (Taye et al., 2024). Many households do not have access to clean potable water, and sanitation levels vary between households.

3.4 Study Population

The original cohort of 1036 students has been described previously (Russel et al., 2023). Of those students, 140 were randomly sampled to be included in the analysis and narrowed down to 138 after exclusion of insufficient samples. A previous test showed no significant difference between characteristics of those included and those excluded from the sample (Taye et al., 2024).

3.5 Measurement and Data Collection

3.5.1 Questionnaire

Each of the children had a guardian fill out a questionnaire form to collect demographic, environmental, lifestyle, clinical, and socioeconomic factors. Demographic questions included sex, age, weight, mode of delivery, etc. Environmental and lifestyle questions included kitchen and house

materials, cleanliness of latrine, bathing location, etc. Clinical factors included vaccination status, current or recent illness, BCG scar, etc. Finally, socioeconomic factors included location of residence, household technology, parental education status, etc.

3.5.2 Fecal sample collection

Fecal samples were initially collected in leak-proof plastic containers before being split in half, one half for detection of parasites, the other for microbiome sequencing. Samples for microbiome sequencing were put in Norgen Stool Nucleic Acid Collection and Preservation Tubes (Norgen Biotek Corp) and then sent to Jimma University where they were frozen within 2 hours. Lastly, those tubes were delivered to Colgate University where they were held at -80°C prior to sequencing.

3.5.3 DNA extraction, library preparation, and sequencing

One smaller portion, up to 250 mg, was taken from the fecal samples to be thawed for extraction. DNA extraction was performed with the DNeasy PowerSoil Pro kit (Qiagen, Germany) according to the manufacturer's instructions. Homogenization was performed using the Beadbeater (BioSpec, Inc. OK, USA) with two 30-second pulses at 4°C. Isolated DNA purity and concentration was tested with the NanoDrop One (ThermoFisher Scientific, MA, USA). DNA integrity was tested by resolving DNA extracts on a 0.8% agarose gel by electrophoresis, afterwards this was visualized with ethidium bromide staining. Microbial DNA extraction was done by PCR amplification of the
16s rRNA V4 variable region using the primers 515F-Y and 806R according to Earth Microbiome Project (EMP) protocols (earthmicrobiome.org/protocols-and-standards/16s). After this initial extraction, DNA samples were stored again at -80°C before being shipped to MR DNA molecular laboratory (MR DNA, TX, USA) for sequencing on the Illumina MiSeq platform following standard Illumina protocol. The raw FASTQ sequence data was demultiplexed using FASTQ Splitter 64 bit v19.07.10.

3.5.4 Amplicon sequencing reads processing

Sequence data was assessed for quality with FastQC (Andrews et al., 2010). Sequences with low quality (quality score < 20) or reads shorter than 50 base pairs were trimmed with Trimmomatic v.39. ASVs were generated using DADA2 (Version 1.26) (Callahan et al., 2016) and R (Version 4.2.2) (Team, 2025) following the 'DADA2 Pipeline Tutorial (1.10) methodology. The software Cutadapt (Martin, 2011) was used to remove adapter sequences. A sample inference algorithm was applied after dereplicating identical reads. Classification of filtered, high-quality sequence reads was performed with the SILVA (SILVAv138) (Quast et al., 2013) reference database after chimeras were removed and paired reads were merged. Contaminant sequences from archaea, mitochondria, and chloroplast were removed before calculating abundance on the newly filtered dataset.

3.6 Statistical analysis

All statistical analysis was done in *R* (version 4.5.1)(Team, 2025). Data was rare-fyed to 20,000 read sampling depth without replacement. Data cleaning and analysis was done with *tidyverse* (Wickham et al., 2019). Plotting of data was done with *ggplot2* (Wickham, 2016). Handling of microbiome features was done with *phyloseq* (McMurdie & Holmes, 2013). When performed, *p*-value adjustments were done according to the Benjamini-Hochberg method for controlling for false discovery rate (Benjamini & Hochberg, 1995). Estimations of alpha diversity were done with the Shannon (Shannon, 1948) and Chao1 (Chao, 1984) biodiversity indices. Tests of alpha diversity were done with Wilcoxon signed-rank tests for equality of medians (Woolson, 2008) in the two sample case. K-sample cases were done with the Kruskal-Wallis test of equality of medians (McKight & Najab, 2010). Effect size calculations showing η^2 were done with the *effectsize R* package (Ben-Shachar et al., 2020). To test differences in gut microbiome composition across groups, also known as beta diversity, we used the Permutational Multivariate Analysis of Variance (PERMANOVA) test from the *vegan* package (Oksanen et al., 2025), plotting the results on a PCA plot with the *MicrobiotaProcess* package (Xu et al., 2023). To test differences in the abundance of genera among differing groups, we used a Wilcoxon signed-rank test for equality of medians (Woolson, 2008) with the Benjamini-Hochberg

method of adjusting *p*-values (Benjamini & Hochberg, 1995). Volcano plots showing effect size and *p*-values across a series of Wilcoxon tests were done with the *Maaslin2* package (Mallick et al., 2020).

3.6.1 Data and Code Availability

All data and code used for analysis of this experiment can be found at https://github.com/MaxWarriner/Microbiome

4 Results

For this study, we performed exploratory analysis to test whether socioeconomic, demographic, environmental, and other lifestyle characteristics impacted the gut microbiome of Ethiopian schoolchildren in Jimma Town, Ethiopia. As a part of a larger study (Taye et al., 2024), a comprehensive survey was completed by a guardian of each child that included various demographic, socioeconomic, lifestyle, and environmental factors about their life. Most children were between the ages of 10-13 (51%), and there were more girls (59%) than boys (41%). Detailed characteristics of the study population can be found in Table 1. A fecal sample taken from each child was sequenced on the V4 region of the 16s rRna gene to profile their gut microbiome composition. From this, we can test if differences from child-to-child affect various measures of the gut microbiome.

4.1 Gut Microbiome Alpha Diversity is Altered by Handwashing Habits, Latrine Use, and Demographics

To test the effects of demographic, socioeconomic, environmental, and lifestyle characteristics on gut microbiome alpha diversity, we performed a series of Wilcoxon Signed Rank tests (2 groups) and Kruskal-Wallis tests (3+ groups) for differences in medians of chao1 and Shannon diversity across groups of Ethiopian school children. We found that alpha diversity was greater in children who reported that they only **sometimes** used soap rather than **always** after using the toilet (Chao1: p = 0.0248, η^2 = 0.032, Shannon: p = 0.0065, η^2 = 0.051) (Figure 1). Furthermore, when comparing children across frequencies of using the school latrine, differing levels of school latrine use had significantly different levels of Chao1 diversity (p = 0.0402, η^2 = 0.033) (Figure 2). Comparisons across Shannon diversity did not meet traditional levels of statistical significance. We also compared

children older than 12 years old to children 12 years old or younger, and found that children older than 12 years old had significantly higher Chao1 diversity (p = 0.0340, η^2 = 0.026) and marginally higher Shannon diversity (p = 0.0806, η^2 = 0.015) than children 12 years old (Figure 3). Lastly, we compared children across two groups of weight. We found that children who weighed 40 kg or more had significantly greater Chao1 diversity (p = 0.0177, η^2 = 0.034) and marginally greater Shannon diversity (p = 0.0836, η^2 = 0.015) than children weighing less than 40 kg (Figure 4).

4.2 Compositional Differences in the Gut Across Kitchen Floor Material Differences

To test multivariate differences in gut microbiome composition across demographic, socioeconomic, environmental, and lifestyle characteristics, we performed a series of PERMANOVA tests using Bray-Curtis and Jaccard distance metrics. We found Beta diversity differences between the groups of children who had cement versus dust kitchen floors among Jaccard (p = 0.003, R^2 = 0.013) and Bray-Curtis (p = 0.007, R^2 = 0.016) distance metrics (Figure 5).

4.3 Hygienic Habits and Home Conditions Enrich Specific Genera

We tested the abundance of genera of bacteria across socioeconomic, environmental, and lifestyle factors with a Wilcoxon signed rank test for equality of medians. We found two differentially abundant genera among children who always used soap after using a latrine/toilet versus children who never/sometimes used soap after using a latrine/toilet: *Elusimicrobium* (β = 3.054, p = 0.0004, q = 0.059) and *Prevotella* (β = 2.491, p = 0.0002, q = 0.059) (Figure 6). We found that three genera were significantly different (p < 0.05) among children who always/sometimes used the school latrine versus those who never did with p < 0.05 and adjusted p < 0.015: *Blautia* (β = 0.671, p = 0.0035, q = 0.143), *Elusimicrobium* (β = −2.821, p = 0.0014, q = 0.108), and *Paludicola* (β = −1.614, p = 0.0026, q = 0.124) (Figure 7). Lastly, we also found differentially abundant genera among children who had cement versus dust/mud kitchen floors: *Lachnoclostridium* (β = −0.811, p = 0.0007, q = 0.097) and *Prevotella_9* (β = 1.621, p = 0.0007, q = 0.097) (Figure 8).

5 Discussion

This cross-sectional study of household and environmental factors that affect the gut microbiomes of Ethiopian children adds novel analysis to an understudied avenue of microbiome research in an understudied population. We observe a novel trend in increasing gut diversity and differing gut composition associated with less traditionally hygienic lifestyle habits and living conditions.

We found statistically significant increases in alpha diversity in children who reported not using soap as frequently as other children after using the latrine (toilet). Handwashing with soap is known to reduce the overall abundance and diversity of microorganisms on the skin microbiome, which can be a positive or negative effect given the types of microorganisms removed (Vandegrift et al., 2017). Pathogens and bacteria are commonly spread via fecal-oral transmission, as the hand microbiome consists of both oral and fecal associated microbes (Shaffer & Lozupone, 2018). Our findings suggest that children who use soap less frequently after latrine use may be exposed to more fecal microbiota on their hands, increasing their overall gut diversity.

We also reported significant increases in the abundance of *Elusimicrobium* & *Prevotella* (before adjustment) for children who use soap less frequently. A small study in humans associated decreases in *Elusimicrobium* with type II diabetes (Almugadam et al., 2020), and overall, *Elusimicrobium* is considered beneficial bacteria in the human gut (Shi et al., 2017), and our analysis shows a consistent low abundance of *Elusimicrobium* in children who always washed their hands with soap after using the toilet. Literature is mixed on the associations with *Prevotella* (Chen et al., 2026). Depending on the study, increased abundance of *Prevotella* is associated with oral inflammation, obesity, and immune diseases. However, decreased abundance of *Prevotella* is also associated with Crohn's disease (Chen et al., 2026). These mixed results highlight the importance of species and strain level analysis in gut microbiome research.

In our sample, we also see a higher frequency of using the school latrine associated with greater intestinal diversity, though only in the Chao1 metric. In a study on schools in Southern Ethiopia, only 7.1% of school latrine floors were considered 'clean,' compared with 41% of home latrines (Grimes et al., 2017). Latrines were also heavily used as there was only around one available per one hundred

children (boys and girls) (Grimes et al., 2017). On the same line of reasoning as previously discussed, handwashing after defecation was only seen in 50% of students (Grimes et al., 2017). The same oral-fecal microorganism pathway could be seen more frequently among students who more frequently use the school latrines, which could be more contaminated with bacteria.

For students with increased school latrine use, also see decreased abundance in *Blautia*, and increased abundance of *Elusimicrobium* & *Paludicola*. Differential abundance results did not survive statistical adjustment. Many studies have been conducted on the link between *Blautia* and obesity, with a meta-analysis concluding mostly inverse correlations and a potential mechanism of *Blautia* in alleviating obesity, but more research is needed (Chanda et al., 2024). Other human studies have shown lower abundances of *Blautia* in people with liver disease (Chen et al., 2021) and Parkinson's (Guo et al., 2022). Lastly, less is known about the *Paludicola* genus. Greater abundance of *Paludicola* in humans has been found to be associated with healthy aging measures (Ramos et al., 2026) and healthy diet (Yang et al., 2023), but also Hidradenitis suppurativa disease (Lelonek et al., 2025) and Kawasaki disease (Yang et al., 2025). In total, the direction of positive/negative health associations with frequency of school latrine usage is mixed in our sample.

Often found among rural families, children living in a house with a dust kitchen floor had significantly different gut compositions than children in a house with a cement kitchen floor, specifically there was found to be a significant decrease in *Lachnoclostridium* & and a significant increase in *Prevotella-9* before adjustment. *Lachnoclostridium* is typically enriched in people with negative health outcomes, such as colorectal polyps (Grion et al., 2024), atherosclerosis (Cai et al., 2022), and adenoma (Liang et al., 2020). For *Prevotella 9*, a reduction in the genera is most often associated with negative health outcomes like type-II diabetes (Wang et al., 2024), Crohn's disease (Hu et al., 2022), Parkinson's disease (Jin et al., 2019), but not always, as in the case of rheumatoid arthritis (Wu et al., 2025). It is surprising to see the differences in abundance across kitchen floor material aligned in their association with negative health outcomes and cement kitchen floors, as those are associated with factors indicating a higher economic status in our dataset, like maternal education status and having a television in the household.

In Ethiopia, almost all rural households, and even half of urban households have earth, sand, or dung floors (Institute & Icf, 2021). There's a lack of research characterizing effects of floor type in Africa on people in the household. Mills et al. find household floor contaminants like animal and pet feces having discernible effects on the gut microbiome (Mills et al., 2023). To our knowledge, there is no current study on the association between the gut microbiome and floor material. Of course, floor material is highly correlated with factors relating to socioeconomic status, lifestyle factors, and other covariates. However, in our analysis, floor material stood out compared to those variables in differences in microbiome diversity and composition.

A limitation of our study is that exploratory observational results are often difficult to reproduce or can be explained by other related covariates. As our data was based on a large survey, along with the high dimensional nature of microbiome studies, this increases the chance of type I errors in analysis, even when our sample size was relatively large. Microbiome studies often struggle to have substantial power given small observable effect sizes, especially when multiple comparison adjustments are made. Even the strongest beta diversity difference we observed had a very small effect size, explaining 1-2% of the multivariate structure. This is why observational (microbiome) studies warrant reproduced outcomes and further investigation for biologically causal analysis.

6 References

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7 Figures

7.1 Alpha Diversity

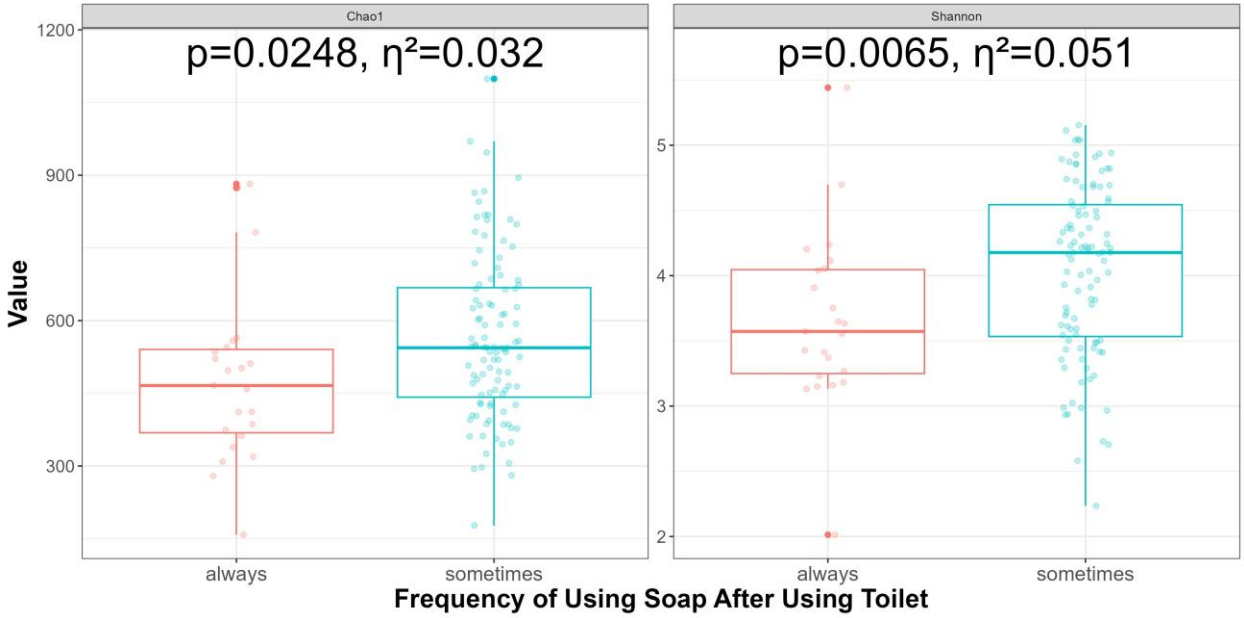


Figure 1: We calculated two metrics of alpha diversity for each child: Shannon and Chao1. The Shannon metric quantifies abundance and evenness of species in an ecosystem as a measure of diversity. Chao1 is an estimate of total species using singletons to infer unobserved species. Hypothesis testing values were done with a Wilcoxon signed rank test for equality of medians, a non-parametric alternative to a two-sample t-test. Across both metrics, the group of children that used soap only sometimes after using a latrine (toilet) had statistically higher levels of alpha diversity for Chao1 diversity (p = 0.0248, η^2 = 0.032) and Shannon diversity (p = 0.0065, η^2 = 0.051)

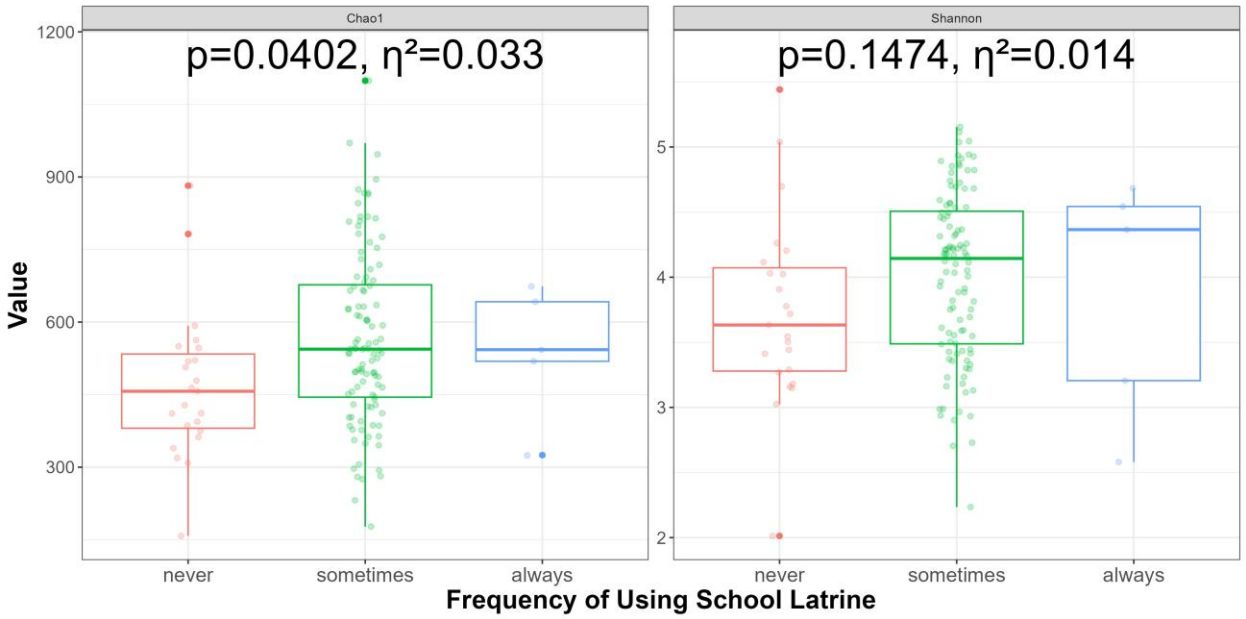


Figure 2: We compared two measures of alpha diversity (Shannon & Chao1) for children across different frequencies of using their school latrine using a Kruskal-Wallis test for equality of medians. Chao1 diversity was significantly different across frequency of using a school latrine (p = 0.0402, η^2 = 0.033). Differences in Shannon diversity did not reach traditional levels of statistical significance.

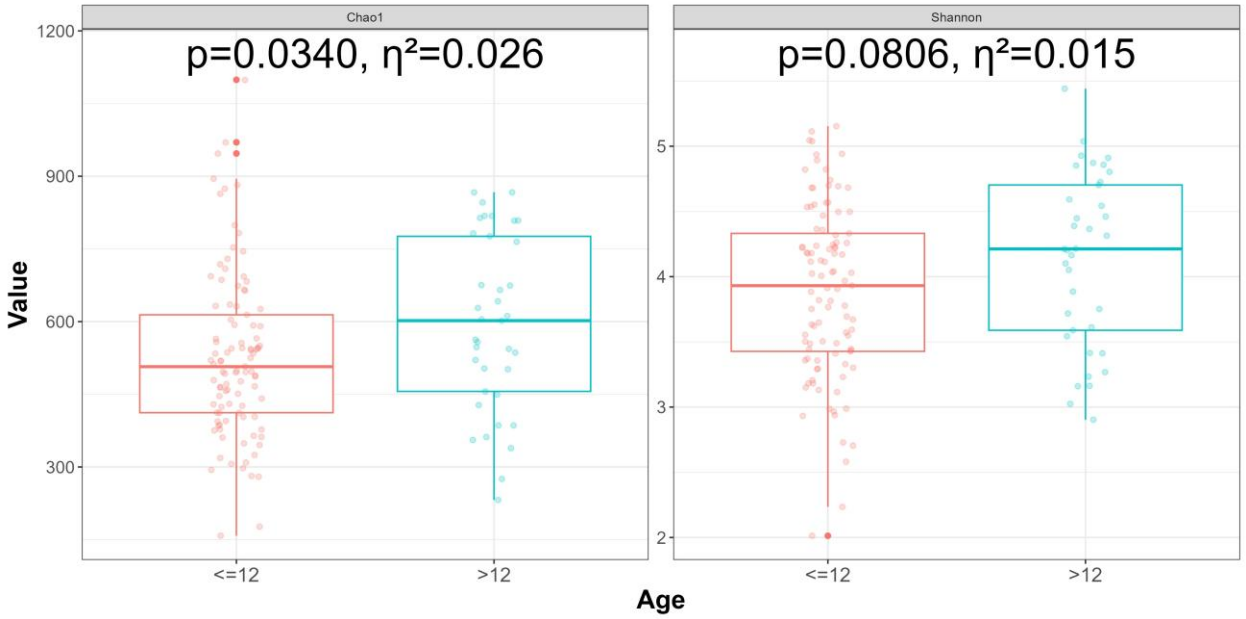


Figure 3: When comparing alpha diversity (Chao1 & Shannon) between children in different age groups, we found in a Wilcoxon signed rank test of equality of medians, that children older than 12 had significantly higher Chao1 diversity (p = 0.0340, η^2 = 0.026) and marginally higher (p = 0.0806, η^2 = 0.015) Shannon diversity.

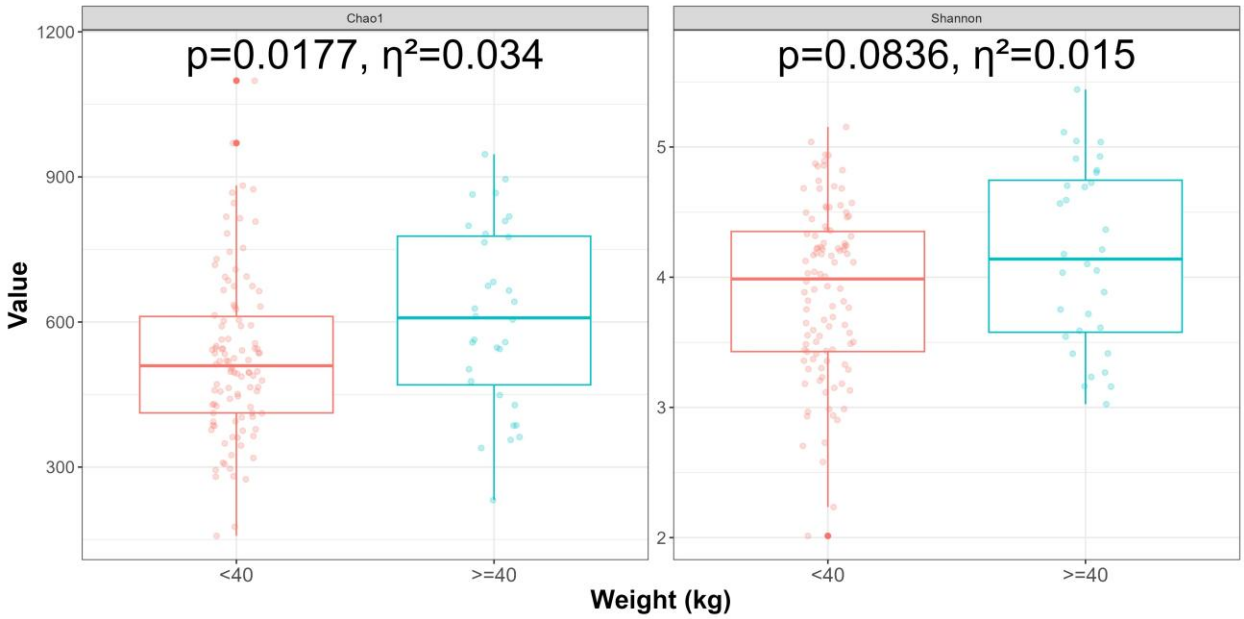


Figure 4: We tested two different measures of alpha diversity (Shannon & Chao1) for children in two different weight groups. Children heavier than or equal to 40 kg had significantly greater Chao1 diversity (p = 0.0177, η^2 = 0.034) and marginally greater Shannon diversity (p = 0.0836, η^2 = 0.015).

7.2 Beta Diversity

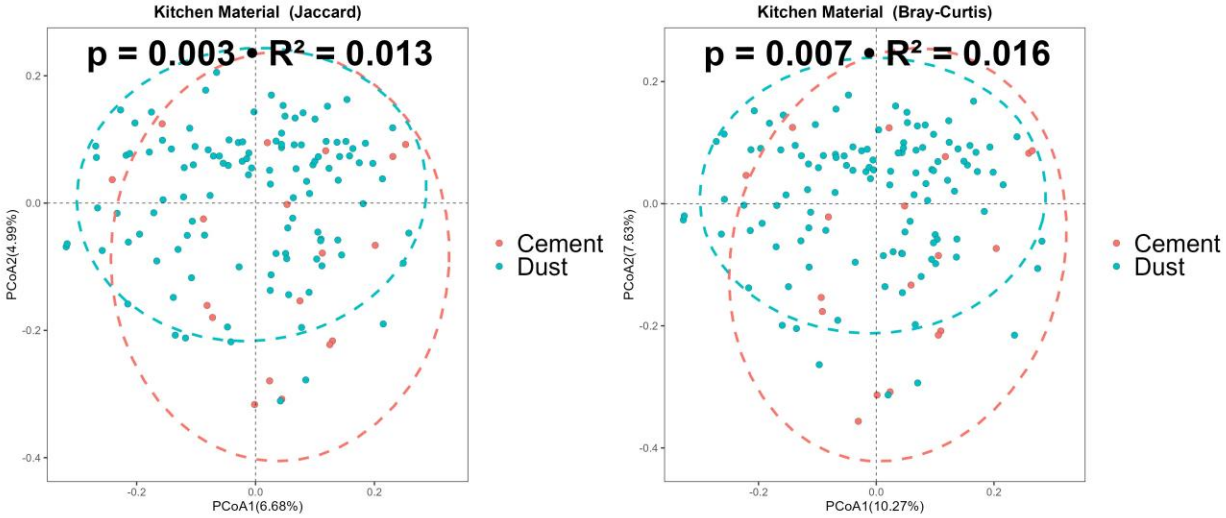


Figure 5: We calculated the beta diversity between groups of children who had kitchens made of dust/mud or cement. Beta diversity measures the difference in species composition between different groups and then plotted each group on a PCoA plot. A PCoA plot uses ordination to place an individual microbiome along two axes, with each axis representing a percentage of variation in diversity accounted for. The groups are then outlined with a 95% confidence interval ellipse. A PERMANOVA test of centroids and dispersion of the ellipses was used to assess differences in beta diversity between the two groups across two distance metrics (Jaccard & Bray-Curtis). Both distance metrics gave statistically significant beta diversity differences between the two groups (Jaccard: $p = 0.003$, $R^2 = 0.013$, Bray-Curtis: $p = 0.007$, $R^2 = 0.016$).

7.3 Differential Abundance

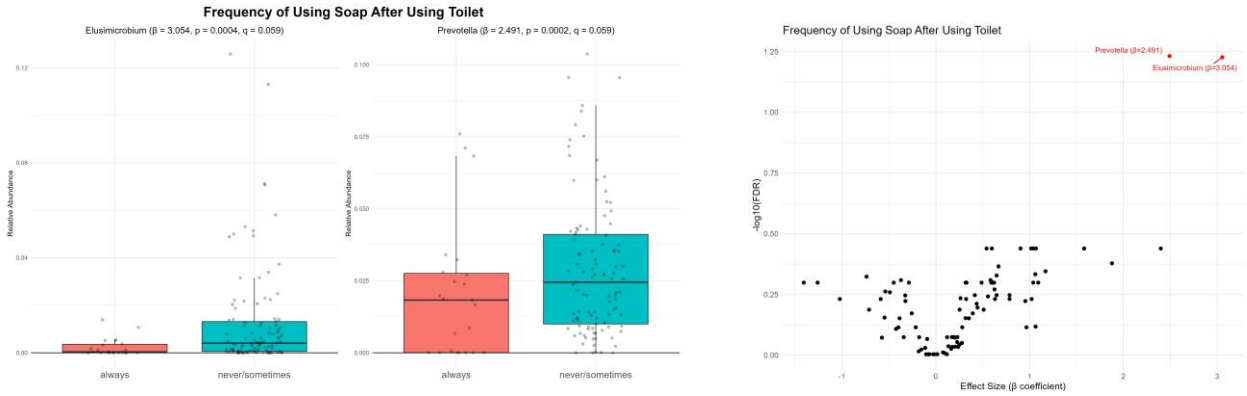


Figure 6: We performed Wilcoxon signed rank tests for equality of medians to test differential abundance of genera for children who always use soap after using a toilet versus children who sometimes/never use soap after using a toilet. Two genera were recognized as having the greatest differential abundance. We found that children who sometimes/never used soap after using a toilet had significantly greater abundance of *Elusimicrobium* ($\beta = 3.054$, $p = 0.0004$, $q = 0.059$) than children who always used soap. Also, children who sometimes/never used soap after using a toilet had significantly greater abundance of *Prevotella* ($\beta = 2.491$ $p = 0.0002$, $q = 0.059$). Both genera were only

marginally significant after Benjamini-Hochberg p-value adjustment. On the right is the two genera highlighted on a volcano plot showing differential abundance effect size (x-axis) and p-value (y-axis, higher = lower p-value).

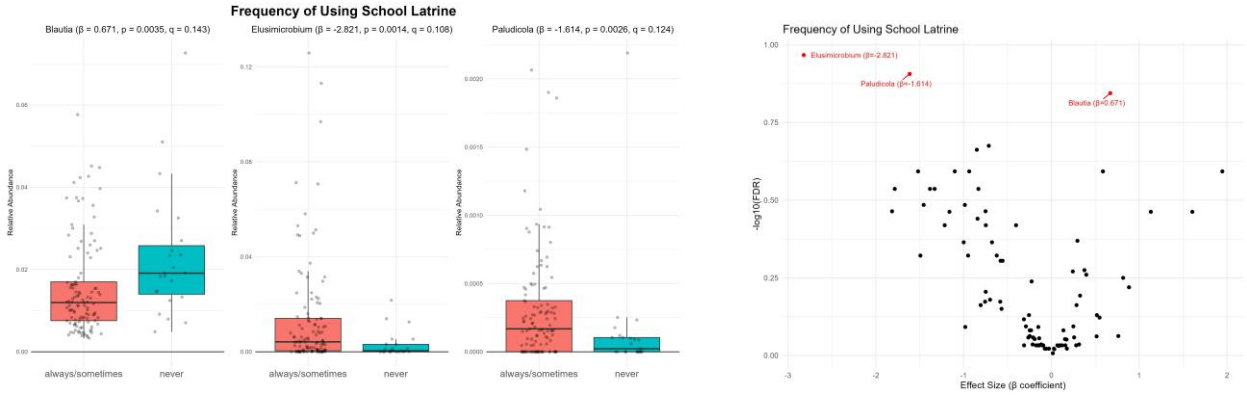


Figure 7: We performed a Wilcoxon signed rank test of equality of medians on the abundance of genera between children who never used the school latrine versus sometimes/always using the school latrine. Genera with Benjamini-Hochberg adjusted p-values < 0.15 were filtered down to 3 differentially abundant genera. Children who never used the school latrine compared to children who sometimes/always used the school latrine had higher abundance of *Bluatia* (β = 0.671, p = 0.003, q = 0.143), lower abundance of *Elusimicrobiom* (β = −2.821, p = 0.001, q = 0.108), and lower abundance of *Paludicola* (β = −1.614, p = 0.003, q = 0.124). None of the values remained statistically significant after adjustment. Also shown on the right is a volcano plot of each genera plotted according to their differential abundance effect size (x-axis), and their p-value (y-axis, higher = lower p-value)

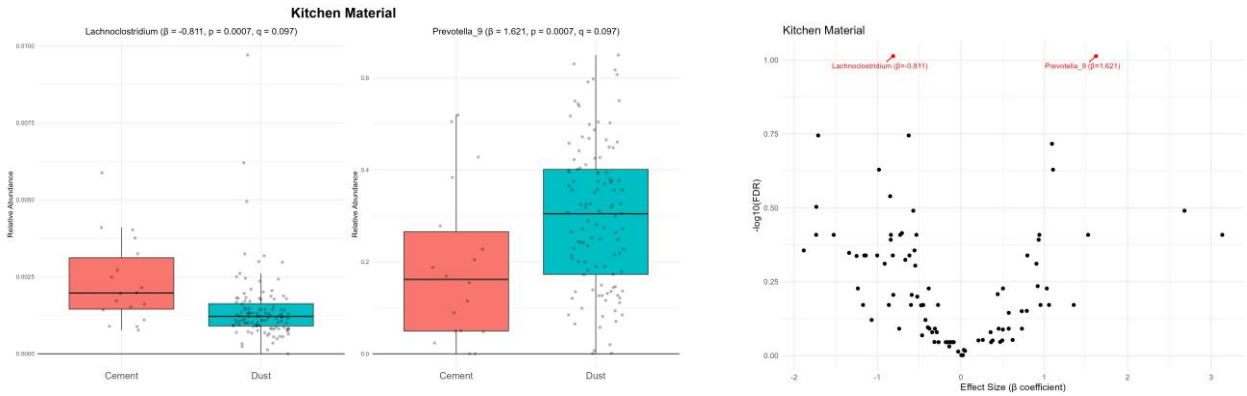


Figure 8: When comparing children with cement versus dust/mud floors in their kitchen, 2 differentially abundant genera were filtered with Benjamini-Hochberg adjusted p-values after performing a Wilcoxon signed rank test for equality of medians. Children with kitchen cement floors had significantly higher abundance of *Lachnoclostridium* (β = −0.811, p = 0.001, q = 0.097) than children with dust/mud kitchen floors. Conversely, children with dust/mud kitchen floors had significantly higher abundance of *Prevotella 9* (β = 1.621 p = 0.001, q = 0.097) than children with cement kitchen floors. Both genera were only *marginally* significant after adjustment. Also on the

right is a volcano plot specifically noting the two differentially abundant genera in comparison to the effect sizes and p-values of the other genera.